



VALYRIAN ASTRONAUTS

WHAT ALREADY EXISTS

WINDOW ALGORITHMS

NASA SLS · Astos ALWA

Ascent-aware windows, but closed and built for US pads

MISSION DESIGN TOOLS

GMAT · STK · Orekit

Orbit propagation; no weather, no launch odds, no web API

LAUNCH WEATHER ODDS

NASA APRA / PACER

Climatology by month and hour; no forecasts, not public



THE GAPS WE FOUND

01

No open orbital analysis of Canso exists

Only environmental and economic reports

02

No forecast-based launch odds tied to the orbit, for any Canadian site

Existing tools stop at climatology

03

The advertised 45.1° LEO orbit was never checked

$\sin \beta = \cos 45.1^\circ / \cos 45.3^\circ = 1.0035 > 1$: no launch heading exists

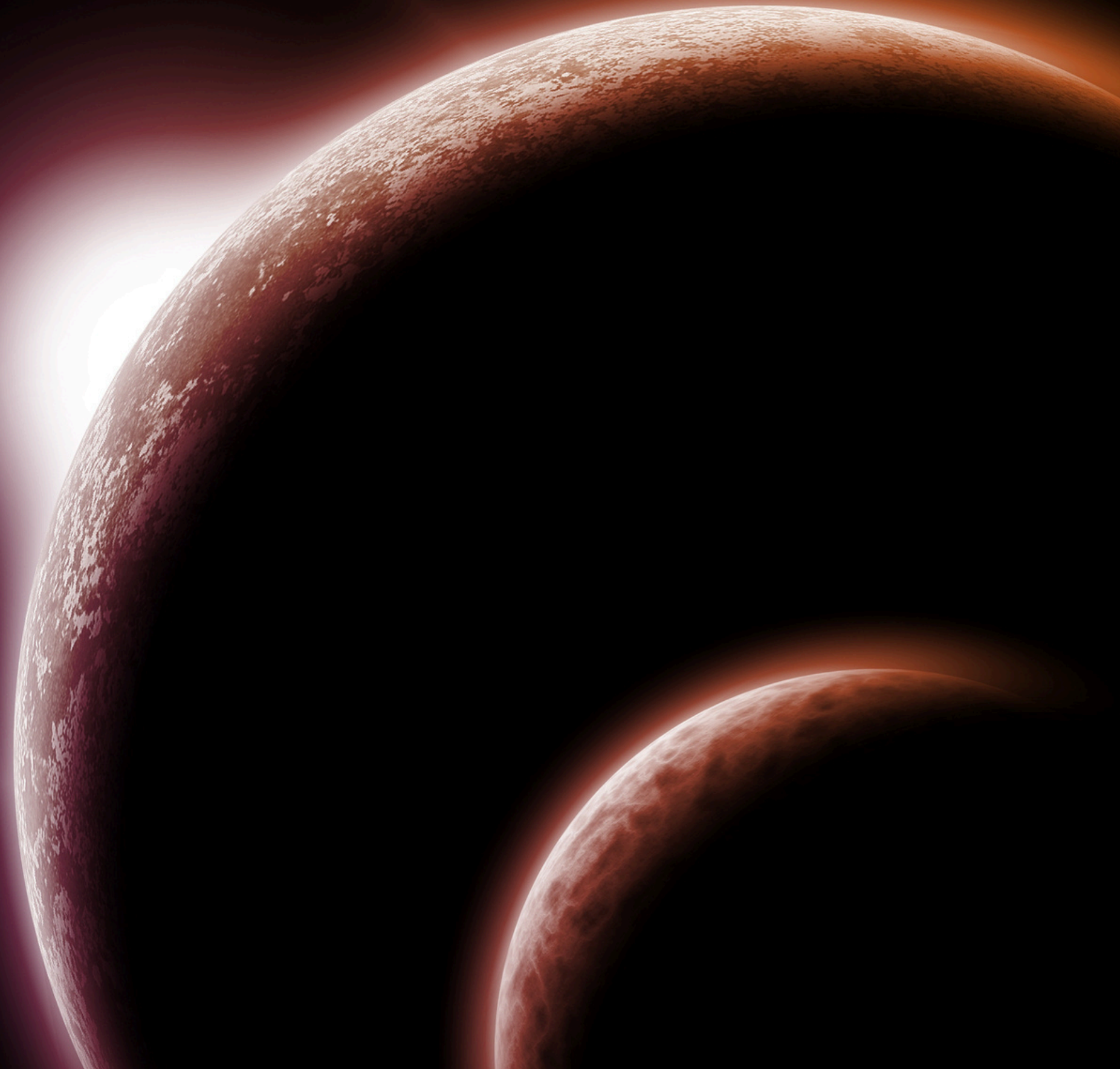


DEMO



HOW THE ENGINE WORKS

- 1 **Reachability** · can a southbound launch reach this tilt?
 $\cos i = \cos \varphi \cdot \sin \beta, \beta \in [115^\circ, 195^\circ]$
else $\Delta v = 2 v_c \cdot \sin(\Delta i / 2)$
- 2 **Window search** · pad rotates under a drifting plane
 $\text{GMST}(t) + \lambda = \Omega(t) + \arcsin(\tan \varphi / \tan i)$
 $d\Omega/dt = -(3/2) J_2 \cdot n \cdot (R/a)^2 \cdot \cos i$
- 3 **Ascent correction** · the ~9 min climb, proven to converge
 $t = \phi(t), |\phi'| \leq 0.017 < 1$ (Banach)
- 4 **Safety screens** · corridor, conjunctions, airspace
debris ellipse: $2000 \text{ km} \cdot \sin 3^\circ \approx 105 \text{ km wide}$
- 5 **Launch odds** · ensemble forecast, scored by hindcast
 $p = P_{wx} \cdot P_{range} \cdot P_{conj}, P_{wx} = (1/N) \sum 1[\text{pass}]$
 $\text{BSS} = 1 - \text{BS} / \text{BS}_{\text{clim}}$



THANK YOU!



— Anand · Het · Rafat · Nafisah · Akash

